Natural Language Processing (AI-3205): Course Overview

This course covers how computers process human language, from basic text cleaning to transformer-based language models.

Assessment breakdown

- Assignments: 20%
- Semester project: 30%
- Midterm exam: 20%
- Final exam: 30%

Semester project

Build a sentiment classifier for Urdu and Roman Urdu product reviews. Teams of three. The project proposal is due in week 6 and the final demo is in week 14. Projects must report precision, recall and F1-score, not only accuracy.

Instructor note: projects that only report accuracy will lose 5 marks, because review datasets are usually imbalanced.

Week 1-2: Text Preprocessing

Tokenization

Tokenization splits text into units such as words or subwords. Modern models use subword tokenization (for example Byte-Pair Encoding), which can handle words never seen during training by breaking them into known pieces.

Stemming vs lemmatization

Stemming chops word endings with simple rules, so 'studies' may become 'studi'. Lemmatization uses vocabulary and grammar to return the dictionary form, so 'studies' becomes 'study'. Lemmatization is slower but more accurate.

Roman Urdu challenges

Roman Urdu has no standard spelling. The same word appears as 'acha', 'achha' and 'accha'. Common fixes are building a spelling normalization dictionary or using character-level and subword models that are less sensitive to spelling variation.

Week 3-4: Word Representations

From one-hot to TF-IDF

One-hot vectors give every word its own dimension, so all words are equally different from each other. TF-IDF weights words by how often they appear in a document, reduced by how common they are across all documents, so words like 'the' get low weight.

Word2Vec

Word2Vec learns dense vectors where words used in similar contexts end up close together. It has two training methods: CBOW predicts a word from its surrounding context, while skip-gram predicts the context from a word. Skip-gram works better for rare words.

These vectors capture relationships, the classic example being: king - man + woman gives a vector close to queen.

Assignment 2 uses skip-gram with a window size of 5 and 100-dimensional vectors.

Week 5-6: Self-Attention and Transformers

Self-attention

In self-attention, every word in a sentence looks at every other word in the same sentence to build a context-aware representation. Each word is projected into three vectors: a query (Q), a key (K) and a value (V).

Scaled dot-product attention

Attention scores are computed as the dot product of a query with every key, divided by sqrt(d_k) to keep values stable, and passed through softmax. The output is the weighted sum of the value vectors.

The transformer

Transformers (Vaswani et al., 2017) drop recurrence entirely and rely on attention, which lets them process all words in parallel. Multi-head attention runs several attention operations side by side so the model can focus on different relationships at once. Because there is no recurrence, positional encodings are added to tell the model the order of words.

Week 7: Language Models and RAG

BERT vs GPT

BERT is an encoder-only model trained to fill in masked words, so it reads text in both directions. It is strong for classification and search. GPT is a decoder-only model trained to predict the next word, which makes it suited to text generation.

Retrieval-augmented generation

Language models only know what was in their training data. Retrieval-augmented generation (RAG) first retrieves relevant passages from a document collection, then gives them to the model together with the question, so answers are grounded in real sources.

Week 7 lab: build a RAG system over the course notes themselves and test it with questions whose answers you already know.